

RESEARCH PAPER

WILEY

Journal of
Biogeography

Assessing the environmental and dispersal controls on *Fagus grandifolia* distributions in the Great Lakes region

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Funding information

University of Wisconsin-Madison Holstrom Environmental Scholarship; National Science Foundation, Grant/Award Number: EF-1065656, EF-1241868

Editor: Ole Vetaas

Abstract

Aim: This paper assesses the relative importance of environmental filtering and dispersal limitations as controls on the western range limit of *Fagus grandifolia*, a common mesic late-successional tree species in the eastern United States. We also test for differences in species–environment relationships between range-edge populations of *F. grandifolia* in eastern Wisconsin and core populations in Michigan. Because environmental conditions between the states differ moderately, while in Michigan dispersal presumably no longer limits *F. grandifolia* distributions, *F. grandifolia* offers a classic case study for biogeographers, foresters, and palaeoecologists interested in understanding processes governing species range limits.

Location: Wisconsin and Michigan, USA.

Taxon: *Fagus grandifolia*.

Methods: This study combines historical datasets of *F. grandifolia* from the Public Land Survey, environmental covariates from soil maps and historical climate data, three spatial scenarios of dispersal limitation, and five species distribution models (SDMs). We test dispersal limitation and environmental filtering hypotheses by assessing SDM transferability between core and edge populations, measuring the importance of dispersal and environmental predictors, and using a residual autocovariate model to test for spatial processes not represented by these predictors.

Results: *Fagus grandifolia* presence was best predicted by total snowfall in Michigan and by dispersal, summer precipitation, and potential evapotranspiration (PET) in Wisconsin. Following the addition of dispersal as a predictor, most Wisconsin models improved and spatial autocorrelation effects largely disappeared. Transferability between core and edge populations was moderate to low.

Main conclusions: Both environmental and dispersal limitations appear to govern the western range limit of *F. grandifolia*. Species–environment relationships differ between range-edge and core populations, suggesting either stronger environmental filtering at the range edge or fine-scale, spatially varying interactions between environmental factors governing moisture availability in core populations. Although lakes, like Lake Michigan, both moderate regional climates and act as dispersal barriers, these effects can be disentangled through the joint analysis of SDMs and historic observational datasets.

KEYWORDSdispersal limitation, environmental filtering, *Fagus grandifolia*, historical ecology, lake-effect snow, Public Land Survey, range limits, spatial autocorrelation, species distribution models

1 | INTRODUCTION

A long-standing question in biogeography is whether contemporary limits of tree species are primarily governed by environmental filtering or by dispersal limitation, and the degree to which current species distributions are out of equilibrium with current environments (Blonder et al., 2017; Davis, 1986; Goring & Williams, 2017; Huntley, Bartlein, & Prentice, 1989; Svenning, Eiserhardt, Normand, Ordonez, & Sandel, 2015; Svenning & Sandel, 2013). A closely related second question is to understand how ecological and environmental processes operating within species ranges vary between edge populations and core populations (Sexton, McIntyre, Angert, & Rice, 2009). To address these questions, we evaluate a classic biogeography case study: understanding the relative controls of climate factors and dispersal limitation on the northwestern range limit of *Fagus grandifolia* (American beech).

In classic niche theory, species distributions are governed by environmental characteristics (Peterson et al., 2011). Proposed environmental filters for plants typically include indices of temperature, moisture availability (Curtis, 1959; Denton & Barnes, 1987; Fang & Lechowicz, 2006; Henne, Hu, & Cleland, 2007; Tardif, Brisson, & Bergeron, 2001), and soil properties that govern moisture and nutrient availability (Beschel, Webber, & Tippet, 1962; Curtis, 1959; Henne et al., 2007; Leak, 1978; Lindsey, Crankshaw, & Qadir, 1965; Maycock, 1963; Carpenter, 1974). Species distribution models (SDMs) formalize the role of environmental filtering by using environmental variables to predict species distributions, which usually requires the assumption that the measured distribution represents the potential range, or fundamental niche (Ozinga et al., 2005; Svenning & Skov, 2004). Species, however, may fail to fully occupy all possible suitable habitats due to dispersal limitations.

Multiple lines of evidence suggest that at least some species distributions are in disequilibrium with the environment; for example, many species ranges have continued to expand since the end of the last glacial period (Svenning & Skov, 2004; Svenning et al., 2015) the alteration of species-climate relationships by recent anthropogenic land use (Goring & Williams, 2017), and evidence for disequilibrium with anthropogenic climate trends in recent decades (Svenning & Sandel, 2013). Inaccuracies in environment-only SDMs and reduction in these inaccuracies when spatial representations of postglacial range expansion are included (Svenning & Skov, 2004) further support dispersal limitation hypotheses. Networks of fossil pollen records show ongoing expansion of several tree taxa into the late Holocene, particularly where large water bodies or other geographical features act as barriers to dispersal (Davis Woods, Webb, & Futyma, 1986; Jackson, 2014; Webb, 1987).

Species dynamics at range margins are further complicated by differential gene flow and selective pressures between range-edge and core populations. Edge populations tend to exhibit reduced fitness (Sexton et al., 2009), reduced genetic diversity (Eckert, Samis, & Loughheed, 2008), and selection for increased dispersal ability (Simmons & Thomas, 2004). Adaptation to environmental stresses can result in intraspecific variation along ecological stress

gradients, such that local adaptations can allow a species to colonize spaces outside the range of the core population (Sexton et al., 2009). These intraspecific variations can reduce predictive accuracy of species distribution models if they assume identical environmental preferences for edge and core populations. However, core and edge populations can be difficult to differentiate along continuous environmental gradients. Conversely, in places where core and edge populations are clearly separated by intervening geographical barriers, these populations can be clearly discriminated, but differences in environmental characteristics between core and edge regions can confound the effects of local adaptation. Large bodies of water, for example, both act as barriers to dispersal gene flow and moderate regional climates. The dual role of these barriers on limiting dispersal and altering regional climates is difficult, although not impossible, to disentangle.

The postglacial expansion of *F. grandifolia* in the Great Lakes region has long been of interest to biogeographers as a case study for testing hypotheses about environmental filtering and dispersal limitation. Paleocological records indicate continuing range expansion through the Holocene, while the modern *F. grandifolia* range boundary in the upper Midwest (Figure 1) is just beyond the western shores of Lake Michigan, which likely both acted as a barrier to dispersal and altered regional climates. During the early Holocene, 8000–7000 years ago, *F. grandifolia* expanded into the Great Lakes region (Figure 1) (Bennett, 1985; Davis et al., 1986; Wang et al., 2015), following the collapse of the Laurentide Ice Sheet and rising temperatures at the end of the last glacial period. Lake Michigan (190 km wide) acted as a barrier to the northwestward expansion of *F. grandifolia*, but approximately 7000 years ago, *F. grandifolia* arrived in eastern Wisconsin (Figure 1), either as a long-distance dispersal event across Lake Michigan (Davis et al., 1986; Webb, 1987) or by expansion around the southern tip of Lake Michigan. A second *F. grandifolia* population expanded north across Lake Superior to the Upper Peninsula of Michigan (Jackson et al., 2014). From there, *F. grandifolia* may have spread southwest into Wisconsin, establishing a stable population approximately 3000 years ago (Davis et al., 1986).

Michigan and Wisconsin, mostly separated by Lake Michigan, have similar geological histories and climates except for differences in wintertime snow accumulation rates (Figure 1), due to the prevalence of lake-effect snow in Michigan (Henne et al., 2007). Lake Michigan additionally moderates regional temperature variations in nearshore regions of both states (Figure 1). Lake Michigan thus provides a clean break between the range-margin populations of *F. grandifolia* in Wisconsin and core populations in Michigan, allowing their environmental and dispersal histories to be separately assessed, yet Lake Michigan also confounds biogeographical interpretations because of its dual roles as dispersal barrier and moderator of regional climates. Multiple prior studies have variously argued that the western range limit of *F. grandifolia* (Bennett, 1985; Davis et al., 1986; Denton & Barnes, 1987; Fang & Lechowicz, 2006; Henne et al., 2007) in the Great Lakes region was primarily affected by environmental or dispersal limitations, with no clear consensus in the prior literature.

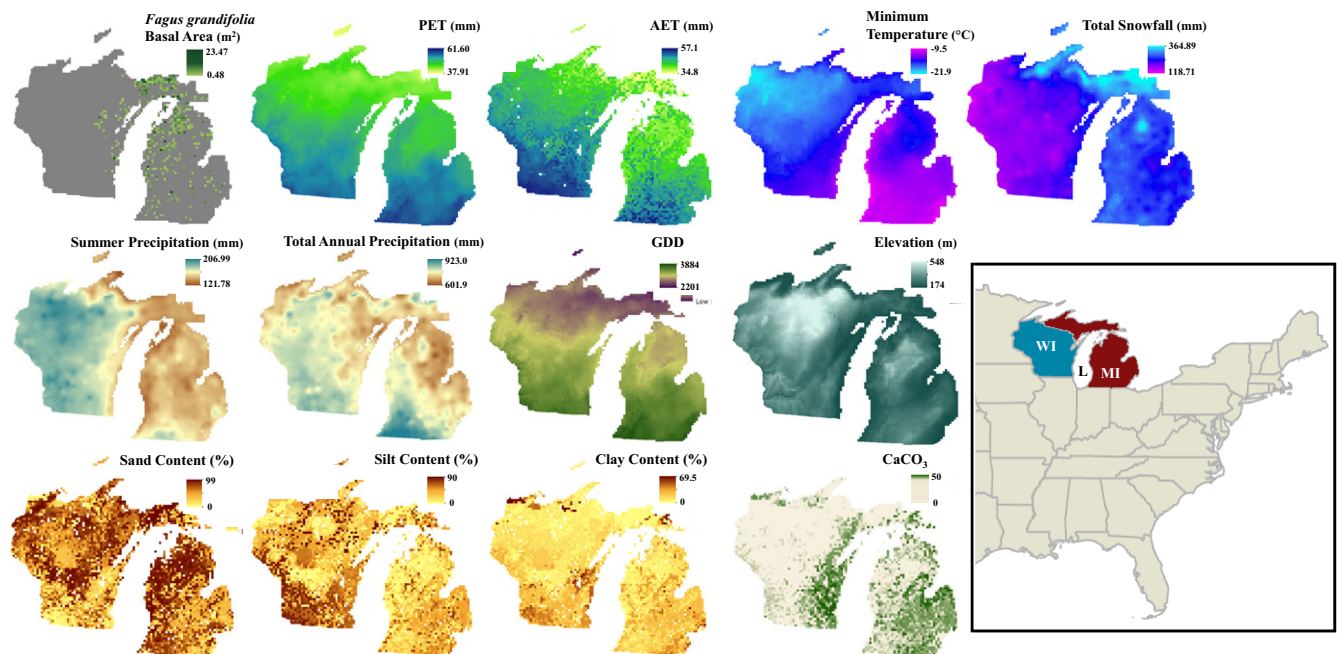


FIGURE 1 Maps of *Fagus grandifolia* basal area at the time of EuroAmerican settlement, environmental variables for Wisconsin and Michigan from PRISM climate data (1895–1924) and SSURGO soil variables, and location of study area within the eastern continental United States (L: Lake Michigan, MI: Michigan, WI: Wisconsin)

This study aims to test hypotheses about the relative importance of dispersal limitations versus environmental filtering as a control on *F. grandifolia* distributions and its western range limit, using a natural comparative experiment between Wisconsin (edge populations) and Michigan (core populations) (Figure 1). We use these two regions to test dispersal limitation hypotheses by building and testing species distribution models (SDMs) for Michigan, where we assume dispersal limitations are minimal by the late Holocene, and Wisconsin, where dispersal limitation may still govern *F. grandifolia* distributions. This research design additionally allows comparison of edge populations in Wisconsin to core populations in Michigan and differences in species–environment relationships between these populations. All SDM experiments rely on the cleaned and standardised Public Land Survey (PLS) dataset by Goring et al. (2016), collocated environmental datasets of historic climates and soil characteristics, four scenarios of dispersal pathways and limitation (in Wisconsin only), and five forms of parametric and machine-learning SDMs. We use multiple lines of inference to assess the relative importance of dispersal limitation and various environmental factors: (a) the intraregional transferability of SDMs separately calibrated in Michigan and Wisconsin, (b) variable selection by SDMs, (c) predictive power of models that incorporate the dispersal scenarios, and (d) tests for sensitivity to spatial autocorrelation (SAC).

2 | MATERIALS AND METHODS

2.1 | Study area

The similar geologic history of Michigan and Wisconsin (Figure 1), combined with contrasting climates and different dispersal histories

of *F. grandifolia*, provides a natural experimental system to compare dispersal limitations and environmental filtering as well as core and edge populations. The legacy of the Laurentide Ice Sheet left many similarities in the states' topography, soils, hydrology, and vegetation (Curtis, 1959), yet it created one feature that separates Wisconsin and Michigan geographically and climactically—Lake Michigan. Wisconsin, west of the lake (Figure 1), receives relatively dry continental air masses. These cold dry air masses receive moisture from Lake Michigan, which then precipitate on the lee side of the lake in western Michigan. This lake-effect snow results in greater wintertime snow accumulation rates in Michigan (Figure 1). Lake Michigan additionally reduces the range of temperature variations in western Michigan and in the easternmost portions of Wisconsin (Figure 1). Lake Michigan also served as a major barrier to the postglacial range expansion of *F. grandifolia* and other thermophilous trees, as discussed above, with *F. grandifolia* arriving first in Michigan and later in Wisconsin (Davis et al., 1986).

2.2 | Species data

This study uses recent compilations of pre-EuroAmerican settlement datasets of *F. grandifolia* compiled, cleaned, and analysed by Goring et al. (2016) to circumvent the strong effects of current and legacy land use on contemporary tree species' distributions and age structures. EuroAmerican land use can obscure the underlying environmental and historical processes governing tree species' distributions and dynamics (Goring et al., 2016; Schulte, Mladenoff, Crow, Merriek, & Cleland, 2007). Conducted in the upper Midwest between 1832 and 1907, the PLS was organized by the General Land Office

to encourage westward expansion, land ownership, and settlement by American settlers (Bourdo, 1956). The Public Land Survey (PLS) provides a rich source of information about the distribution of tree species in the central and western United States just prior to the rapid intensification of land use by EuroAmerican settlers (Goring et al., 2016; Liu, Mladenoff, Keuler, & Moore, 2011; Schulte & Mladenoff, 2001).

Surveyors documented the species, diameter, azimuth, and distance to witness trees identified at the corners of a 1-mile grid system (Stewart, 1935). The Goring et al. (2016) compilation draws on prior digitizations of PLS surveys for Wisconsin (Maines & Mladenoff, 2000; Schulte, Mladenoff, & Nordheim, 2002) and Michigan (USFS-NRCS, 1995); these data were further cleaned and standardized to reduce surveyor bias and transcription error (Goring et al., 2016). For example, all but the two closest witness trees to each corner point were removed, as were line and meander trees. To reduce transcription errors, outlier data points (e.g., trees with diameter at breast height >254 cm) and discrepancies in the digitized data (e.g., a tree with a recorded azimuth despite being at plot centre) were excluded. Other modifications include corrections for azimuthal censoring and survey sampling design. The final dataset was reprojected into Albers Conic Equation and aggregated into 8-km grids displaying mean basal area ($\text{m}^2 \text{ha}^{-1}$) of each recorded species (Goring et al., 2016). From this dataset, we created an 8-km gridded presence/absence dataset of *F. grandifolia* across Michigan and Wisconsin.

2.3 | Environmental data

To assess the effects of physical environmental factors on *F. grandifolia* presence, we developed 12 environmental datasets for Wisconsin and Michigan (Figure 1). These datasets represent soil, elevation, and early 20th century climate variables pertinent to tree establishment and survival. To match the PLS *F. grandifolia* data, each environmental dataset was reprojected to Albers Conic Equation and aggregated via bilinear interpolation into the same 8-km grid as the Goring et al. (2016) PLS data.

Soil property datasets were acquired from the Natural Resource Conservation Service (NRCS) Soil Survey Geographic Database SSURGO (<https://datagateway.nrcs.usda.gov/>), using the NRCS Soil Data Development Toolbox in ArcGIS. Soil properties included soil texture (sand, silt, and clay content), chemistry (CaCO_3), and water-holding capacity (WHC) (Figure 1). Each soil variable was compiled by calculating a weighted average of all soil horizons within the top 150 cm.

A 30-year (1895–1924) average historic climate of the upper Midwest was reconstructed using PRISM data (PRISM Climate Group, Oregon State University, <http://prism.oregonstate.edu/historical/>, created Aug 2013) (Figure 1). The PRISM data are a gridded climate data product based on historical meteorological stations and climactically aided interpolation. PRISM offers monthly historic US climate datasets at a 4-km resolution beginning at 1895 CE. From the PRISM historic climate data, we calculated 30-year climatic

means of standard climatic and bioclimatic variables: minimum temperature, total precipitation (mm/yr), summer (July–August) precipitation, potential evapotranspiration (PET), AET, growing-degree days (GDD; base 0°C), and total snowfall (mm/yr). We used modified Thornthwaite equations compiled and revised by Lutz, Wagtendonk, and Franklin (2010) to calculate PET, AET, GDD, and total snowfall. Calculation of all variables except AET relied solely on PRISM data; AET also required SSURGO WHC data. Total PET and AET were calculated using average monthly precipitation and temperature values. Minimum temperature was calculated as an average of the annual daily minima across the 30 years. We summed the 30-year average precipitation for July through August to compute summer precipitation. To calculate total snowfall, we summed precipitation when temperatures dropped below freezing. Snowfall is reported in water depth; SDM analyses are unaffected by choice of snowfall depth units.

Digital elevation models (DEM) were acquired from the Wisconsin Department of Natural Resources (https://p.widencdn.net/tebltd/DEM_30m, created Jan 2007) and the Michigan Department of Technology, Management, and Budget (https://www.dnr.state.mi.us/spatialdatalibrary/sdl2/Topography/DEM/_Statewide/Michigan_DEM_250k_90m.exe). The native resolutions of the Wisconsin and Michigan elevation datasets were 30 and 90 m respectively.

2.4 | Dispersal

To represent the hypothesis that Wisconsin *F. grandifolia* populations are in part controlled by dispersal limitation, we used spatial processes based on palynological evidence that *F. grandifolia* entered eastern Wisconsin during the middle to late Holocene.

Because the exact initial entry point is uncertain, we created three alternative dispersal scenarios, representing entries from the Upper Peninsula of Michigan (north), Illinois (south), or long-distance dispersal across Lake Michigan (east) (Davis et al., 1986). Each dispersal scenario was represented by a map of Euclidean distances to the hypothesized entry point (Figure 2), and an additional dataset represented all three scenarios (All). We analysed Euclidean distance instead of effective distance (Franklin, 2010; Rogers, Jantz, & Goetz, 2017) because topographic complexity in Wisconsin is low, so few regional barriers to dispersal exist beyond the Great Lakes themselves. Our analysis of dispersal limitation comprises only Wisconsin, as dispersal was assumed to not limit *F. grandifolia* distributions in Michigan-based models.

2.5 | Variable selection

The SDM prediction accuracy can be confounded by high correlation among predictor variables, so SDM analyses typically reduce environmental variables to decrease correlation (Franklin, 2010). Variable selection was performed through a priori reasoning about ecological processes and an evaluation of the correlation structure among predictor variables using the “corplot” R package (Taiyun & Viliam, 2016) (Figure 3). In this preanalysis, *F. grandifolia* basal area ($\text{m}^2 \text{ha}^{-1}$) was

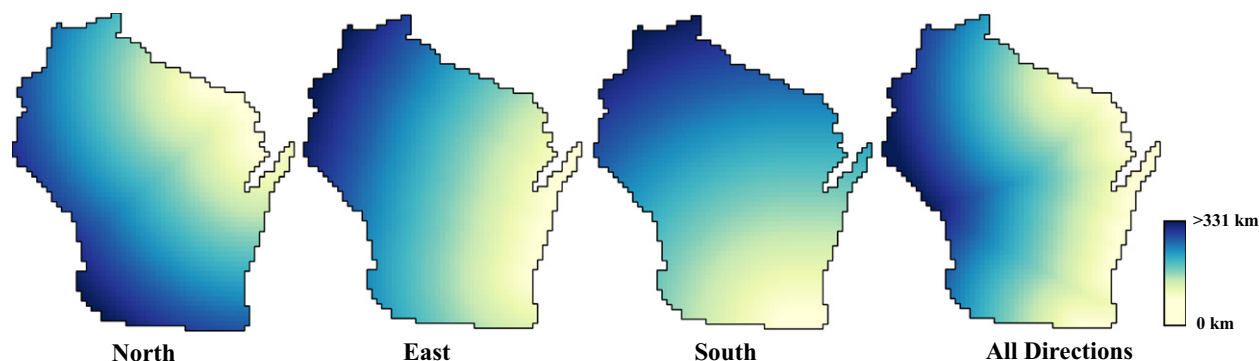


FIGURE 2 Spatial processes representing distance to *Fagus grandifolia* entry into Wisconsin, for four scenarios representing different locations of entry

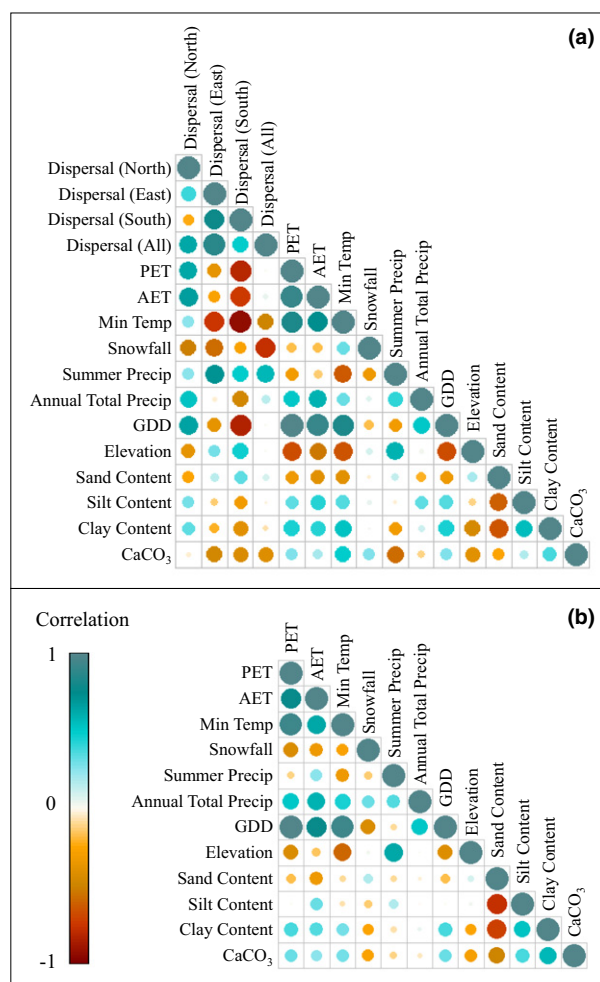


FIGURE 3 Correlation plot for the environmental and dispersal variables for a) Wisconsin edge populations and b) Michigan core populations of *Fagus grandifolia*

used where *F. grandifolia* was present, and Wisconsin and Michigan data were analysed separately.

From the 12 variables assessed (Figure 1), six were chosen for model development. We chose sand content to represent soil texture, for consistency with prior studies on the climatic and edaphic

controls on *F. grandifolia* presence (Beschel et al., 1962; Curtis, 1959; Henne et al., 2007; Leak, 1978; Maycock, 1963). We included calcium as its distribution correlates with that of *F. grandifolia* in Wisconsin (Curtis, 1959). PET, AET, minimum temperature, and GDD also showed a high correlation. Of these variables, we used PET, because it tends to be the most often represented in literature concerning *F. grandifolia* distribution (Denton & Barnes, 1987; Fang & Lechowicz, 2006; Henne et al., 2007). To address seasonality of precipitation (Denton & Barnes, 1987; Henne et al., 2007; Tardif et al., 2001) and hypotheses about lake-effect snow controls on *F. grandifolia* distributions (Henne et al., 2007), we used both total snowfall and summer precipitation. After variable selection, our set of environmental and dispersal variables used in SDM modelling consisted of sand content, calcium carbonate content, PET, total snowfall, summer precipitation (Figure 1), and (for Wisconsin only) the four dispersal scenarios (Figure 2).

2.6 | Species distribution modelling experimental design and evaluation

All SDM experiments were run using the “sdm” R package (Naimi & Araujo, 2016) and the presence/absence *F. grandifolia* data. Experiments were run separately for Wisconsin and Michigan to test the transferability of SDMs calibrated separately for Wisconsin and Michigan *F. grandifolia* range-margin and core populations. Seven SDM calibrations were run: (a) SDMs calibrated using environmental predictor variables from Michigan, (b) SDMs calibrated using environmental Wisconsin predictor variables without a dispersal spatial process, (c–f) Wisconsin environmental predictor variables with a dispersal spatial variable included (all, north, east, south), and (g) using solely the dispersal spatial variable (east scenario) and no environmental variables. Models were calibrated using half the dataset (randomly assigned); the other half was used to validate prediction data. The first two calibrations were used to predict distributions in Wisconsin and Michigan; calibrations with dispersal-related spatial predictors were used for Wisconsin only. Our tests of hypotheses about the environmental and historical controls on *F. grandifolia* in Wisconsin thus rely on four lines of inference: (a) the weighted importance assigned to individual environmental and spatial variables in the SDMs, (b) the transferability of models between

Wisconsin and Michigan, (c) the predictive power of Wisconsin models that include dispersal, and (d) tests for spatial autocorrelation (below).

We include multiple models to reduce dependency of inferences on model selection, as no single kind of SDM performs best for all species (Elith & Graham, 2009). These models include two parametric modelling techniques, generalized linear models (GLM) and generalized additive models (GAM), and three machine learning models, random forests (RF), boosted regression trees (BRT), and support vector machine (SVM). GLM is a multiple linear regression that allows for non-normal error distributions; GAM is a modified GLM that attempts to account for varying relationships within the data by including a spline function. BRT and RF are decision-tree methods that create nodes to represent attributes and branches for each decision outcome. These classification trees form forests that the model refines; methods for growing and pruning trees differ for RF and BRT. SVM predicts binomial outcomes by maximizing separation of training data along multiple axes (based on the predictor values) and subsequently drawing a line between categories.

Model performance was assessed by evaluating the ability to predict *F. grandifolia* presence, using the "sdm" R package. We used two metrics to model accuracy—area under the curve (AUC) (Huang & Ling, 2005) and true skill statistic (TSS) (Allouche, Tsoar, & Kadmon, 2006). AUC represents the area under a receiver operating characteristic (ROC) curve, which plots the true positive rate against the false positive rate for all thresholds and was calculated using the "dismo" R package (Hijmans, Phillips, Leathwick, & Elith, 2017). TSS is a modified kappa statistic, which compares the classification score against a random classification; TSS values reported are the maximum TSS across all thresholds, as calculated by the "ecospat" R package (Broennimann, Di Cola, & Guisan, 2018). We used the SDMs to predict the likelihood of *F. grandifolia* presence in Wisconsin and Michigan (Table 1). For

each model, we determined the relative importance of each prediction variable and evaluated variable rankings in terms of model performance. A high ranking of a variable as a predictor, combined with a high goodness-of-fit for the model prediction, indicates importance of that variable in determining *F. grandifolia* distribution.

2.7 | Spatial autocorrelation

Spatial autocorrelation (SAC) is common in ecological data and, if not accounted for, can exaggerate the importance of other explanatory variables (Crane, Liedloff, & Wintle, 2012; Dormann et al., 2007; Guisan & Thuiller, 2005). We used a residual autocovariate model (RAC) (Crane et al., 2012) to assess potential influences of SAC on our results and test whether adding dispersal distance as an explanatory variable removed those effects. The RAC is an extension of the autologistic model, which uses the autocovariate between each location and its neighbours to correct for SAC. The RAC extension further reduces autocorrelation biases by also including an autocovariate for residuals of the autologistic model (Crane et al., 2012). We created a RAC extension for the GLM, one of the best-performing SDMs, for the MI-MI, WI-WI without Dispersal, and WI-WI with Dispersal (east) experiments and compared goodness-of-fit and variable rankings of the RAC and GLM experiments (Table 1). High goodness-of-fit values of RAC relative to GLM indicate that there is a spatial process unaccounted for in the GLM experiments (Guisan & Thuiller, 2005).

3 | RESULTS

3.1 | Correlational structure between *F. grandifolia* and predictor variables

Correlations of *F. grandifolia* basal area to environmental variables differed between Wisconsin range-edge and Michigan core populations (Figure 4), with a few exceptions. In both regions, *F. grandifolia* positively correlated with snowfall (>0.244) and total precipitation and negatively correlated with summer precipitation (<-0.105). For some environmental variables, the directionality of relationships was opposite between regions. For example, sand negatively correlated with *F. grandifolia* basal area in Wisconsin (-0.290), but positively correlated in Michigan (0.120). Dispersal, represented as distance to *F. grandifolia* entry into Wisconsin, negatively correlated with *F. grandifolia* abundance (<-0.122 , a negative correlation is consistent with dispersal limitation) for all dispersal scenarios except for the North scenario (0.186).

3.2 | SDM results: Goodness-of-fit (AUC/TSS)

Overall, the Wisconsin models had a higher goodness-of-fit, for both the AUC and TSS, than the Michigan models (Figure 5), suggesting that *F. grandifolia* is more strongly constrained by environmental and dispersal factors in the range-margin populations of Wisconsin than in Michigan. Transferability of SDMs in the MI-WI experiments was generally moderate to low: average AUC/TSS values for the Michigan

TABLE 1 Model calibration and prediction naming conventions

Name	Calibration data	Predicted Region
MI-MI	Michigan	Michigan
MI-WI	Michigan	Wisconsin
WI-WI	Wisconsin	Wisconsin
With Dispersal (All)	With Dispersal (All)	
WI-WI	Wisconsin	Wisconsin
With Dispersal (north)	With Dispersal (north)	
WI-WI	Wisconsin	Wisconsin
With Dispersal (east)	With Dispersal (east)	
WI-WI	Wisconsin	Wisconsin
Only Dispersal (east)	Only Dispersal (east)	
WI-WI	Wisconsin	Wisconsin
With Dispersal (south)	With Dispersal (south)	
WI-WI	Wisconsin	Wisconsin
Without Dispersal	Without Dispersal	
WI-MI	Wisconsin	Michigan
Without Dispersal	Without Dispersal	

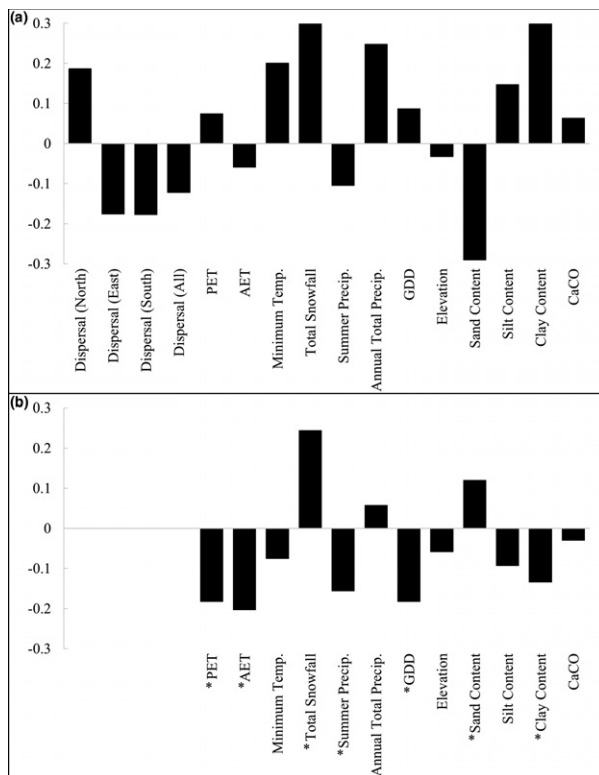


FIGURE 4 Correlation between *Fagus grandifolia* basal area (m^2) and predictor variables for a) Wisconsin ($df = 30$) and b) Michigan ($df = 351$). * Indicates significance at the $p < 0.05$ level

models predicting *F. grandifolia* distribution in Michigan and Wisconsin were 0.728/0.340 and 0.576/0.318, respectively, and most MI-WI experiments substantially overpredicted the distribution of *F. grandifolia* (Figure 6). The MI-WI experiments had the greatest range in performance among models (0.258/0.014 to 0.911/0.6689 AUC/TSS) of all the experiments, suggesting high sensitivity to choice of model. Similarly, the Wisconsin-calibrated SDMs (without Dispersal) had modest transferability, with an average AUC/TSS of 0.842/0.596 and 0.620/0.181 when predicting *F. grandifolia* presence in Wisconsin and Michigan, respectively. Goodness-of-fit also varied widely among models. The poorer performance of the MI-WI and WI-MI scenarios than the MI-MI and WI-WI scenarios (Figure 5) indicates only partial transferability in species–environment relationships between the range-margin and core populations.

The SDM goodness-of-fit improved when dispersal (Figure 2) was included for all scenarios (Figure 5); both TSS and AUC ranked the North scenario as the best-performing scenario (0.714 mean TSS and 0.905 mean AUC). However, when dispersal distance (east entry point) was the only predictor, mean goodness-of-fit decreased substantially (0.690/0.567 AUC/TSS), suggesting that WI distributions may be governed by both environmental and dispersal factors.

Across all experiments, the GAM and GLM SDMs had the highest AUC and TSS scores, respectively (Figure 5). Only for the WI-WI South Dispersal and WI-WI No Dispersal scenarios were these models outperformed by the RF SDM.

3.3 | SDM results: Variable importance

Variable importance differed between the Michigan core population and Wisconsin range-margin populations (Table 2). Models calibrated with Michigan data ranked summer precipitation, total snowfall, sand content, and PET highly, while models calibrated only with Wisconsin environmental data universally ranked summer precipitation as the topmost variable and PET, snowfall, and CaCO_3 as important variables, while universally ranking sand content low and often ranking total snowfall low.

When dispersal was included in the WI experiments, it became the topmost predictor (east entry point) or second-most important predictor (other dispersal scenarios), with summer precipitation usually the other most important variable. Sand content received little weight. When dispersal was represented as originating from across Lake Michigan (east), four of five models ranked dispersal as the most important variable. Throughout all models, PET often ranked in the middle, suggesting that temperature is generally of secondary importance at this spatial scale.

3.4 | Testing for spatial autocorrelation: GLM and RAC analyses

The GLM and RAC analyses indicate that there is residual spatial autocorrelation in the Wisconsin simulations of *F. grandifolia* when dispersal is not accounted for, based on the large goodness-of-fit increase between the GLM and RAC models (Table 3, WI-WI without Dispersal). However, the difference in goodness-of-fit between the GLM and RAC models largely disappears when dispersal is included as a spatial process, indicating that including dispersal to entry point largely removes residual SAC (Table 3, WI-WI with Dispersal). The MI-MI experiment showed a small improvement in goodness-of-fit between the GLM and RAC experiments ($<0.029/0.028$ AUC/TSS), suggesting that its *F. grandifolia* populations are modestly affected by dispersal limitation or some other unexplained spatial process. Variable rankings were largely unchanged between the GLM and RAC experiments, with some pairwise swaps, for example, between summer precipitation and total snowfall in the MI-MI model and between the bottom-ranking variables CaCO_3 , total snowfall, and sand content in the WI-WI without Dispersal model.

4 | DISCUSSION

4.1 | Assessing the relative importance of environmental and dispersal controls on *F. grandifolia*

Species distributions are limited by three sets of factors: environmental filtering, dispersal limitation, and species interactions (Peterson et al., 2011). Most biogeographical research seeks to understand the relative influence of these factors on current and historic distributions, with particular concern about whether current distributions of species are in increasing disequilibrium with climate, due to the rapid environmental changes of the last several centuries (Blonder et al., 2017; Goring &

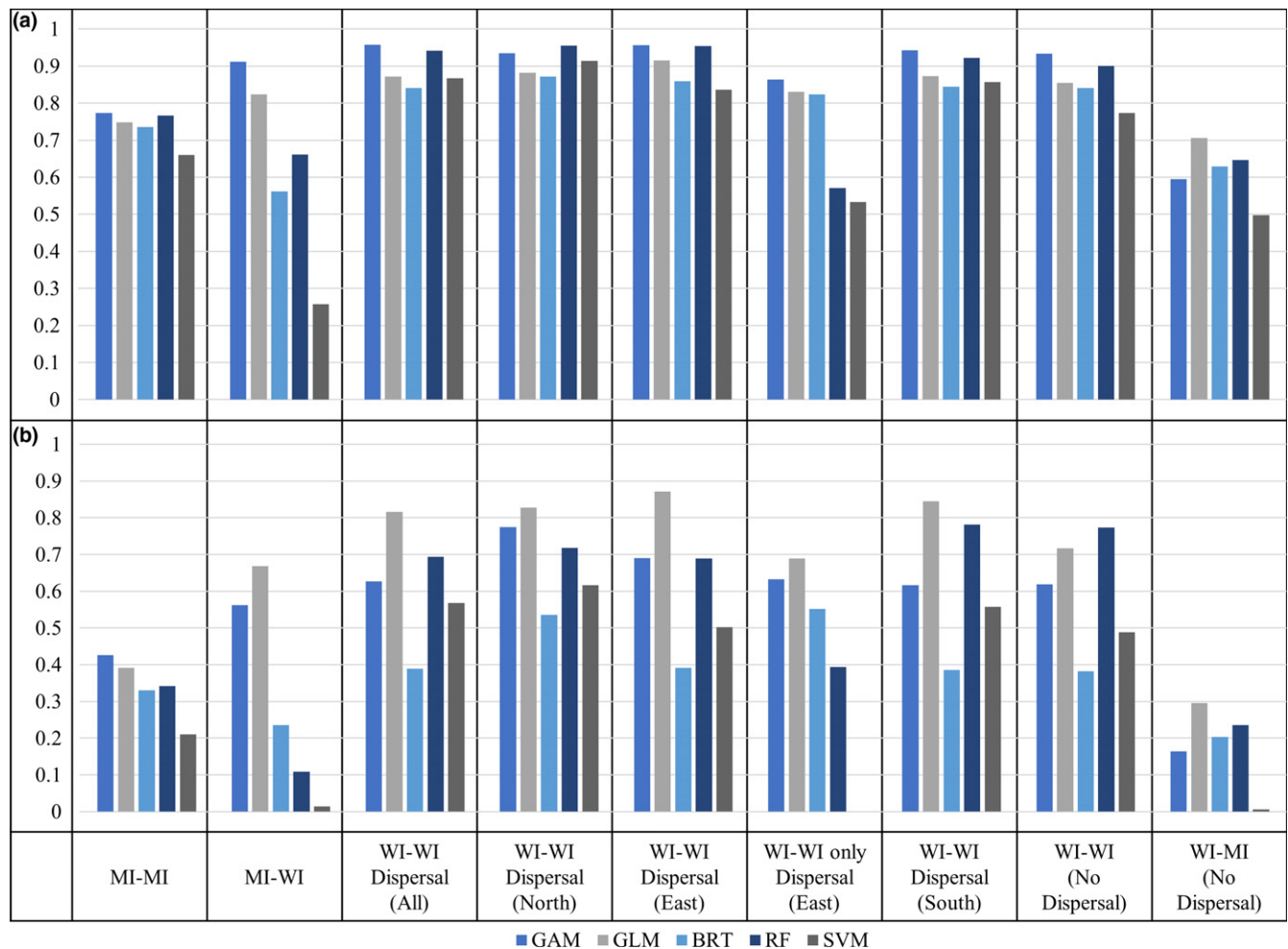


FIGURE 5 Goodness-of-fit for all SDM experiments, represented by (a) area under the curve (AUC) and (b) true skill statistic (TSS). Bar colours represent SDMs. The WI-WI Dispersal-only (east) SVM experiment did not yield a maximum TSS as the predicted probabilities were within 0.01

Williams, 2017; Svenning et al., 2015). Our study supports the hypothesis that the western range limit of *F. grandifolia* is at least partially dispersal-limited, although its distribution is also governed by environmental factors. Multiple lines of inference support the role of dispersal limitation. First, adding distance to *F. grandifolia* entry into Wisconsin as a proxy for dispersal limitation improved model prediction of *F. grandifolia* distributions, because goodness-of-fit for the Wisconsin experiments was highest for experiments that included dispersal (Figure 5). Second, Wisconsin SDMs consistently ranked dispersal as the most or second-most important factor when included (Table 2). The selection of summer precipitation and dispersal as important predictors of *F. grandifolia* distributions in Wisconsin (Table 2) is consistent with the spatial patterns of these variables in Wisconsin in that they closely match the spatial distribution of *F. grandifolia* (Figure 1). Notably, most SDMs select both variables as having predictive power (Table 2), suggesting that neither summer precipitation nor dispersal limitation alone can fully explain *F. grandifolia* distributions in Wisconsin.

Third, the RAC analyses indicate that the spatial process used here to model dispersal largely accounted for SAC effects present in WI-WI

without Dispersal models (Table 3). Improvement of GLMs following addition of the RAC extension is indicative of residual SAC, which can signal that important spatially autocorrelated predictors have been excluded (Guisan & Thuiller, 2005). For the WI-WI without Dispersal experiment, GLM AUC and TSS improved with the RAC extension more than for the Wisconsin with Dispersal (east) model (Table 3). The Michigan model likewise improved marginally when RAC was included, indicating that the lower performance of the Michigan models was not a result of SAC. For all scenarios, variable ranking changed little following RAC analyses (Table 4). These RAC analyses thus suggest that dispersal limitation, represented here as distance to potential entry points into Wisconsin, is a spatial process that partially governs *F. grandifolia* distributions. Our approach to using SAC as an indicator of explained spatial structure caused by dispersal limitation is similar to using Moran's *I* to map spatial patterns in model residuals (Dormann et al., 2007), because RAC is an autologistic model that calculates autocovariates from model residuals. Our approach then uses these residuals to directly assess their effect on model performance (Crane et al., 2012).

Within the broader study of controls on species distributions, the *F. grandifolia* case study is interesting because the key barrier to

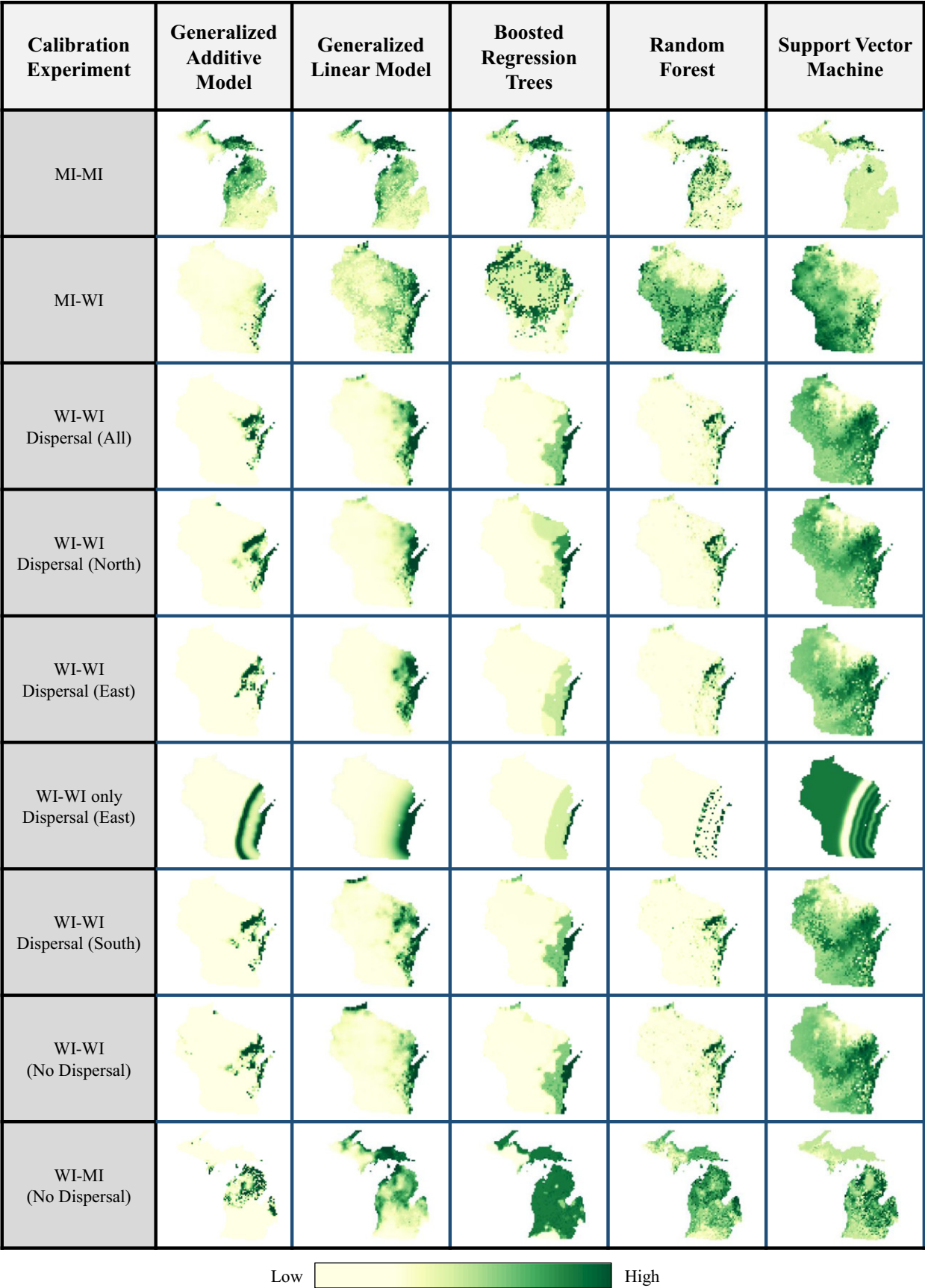


FIGURE 6 Mapped probabilities of *Fagus grandifolia* presence in Wisconsin and Michigan for each combination of calibration experiment and SDM

TABLE 2 Variable ranking for each SDM calibration experiment

Calibration Experiment	GAM	GLM	BRT	RF	SVM
Michigan	Snowfall	Summer Precip.	Sand	PET	Snowfall
	0.355	0.341	0.283	0.295	0.789
	Summer Precip.	Snowfall	Summer Precip.	Summer Precip.	Sand
	0.229	0.247	0.238	0.287	0.230
	PET	Sand	Snowfall	Snowfall	PET
	0.211	0.225	0.223	0.204	0.179
Wisconsin with Dispersal (All)	CaCO ₃	PET	PET	Sand	CaCO ₃
	0.149	0.045	0.0212	0.196	0.161
	Sand	CaCO ₃	CaCO ₃	CaCO ₃	Summer Precip.
	0.086	0.014	0.000	0.153	0.056
	PET	Summer Precip.	Summer Precip.	Summer Precip.	Summer Precip.
	0.777	0.843	0.820	0.531	0.638
	Dispersal	Dispersal	Dispersal	Dispersal	PET
	0.600	0.501	0.041	0.409	0.388
	Summer Precip.	PET	Snowfall	PET	Snowfall
	0.317	0.402	0.010	0.260	0.339
	CaCO ₃	Snowfall	CaCO ₃	Snowfall	CaCO ₃
	0.268	0.115	0.002	0.121	0.228
Wisconsin with Dispersal (north)	Sand	CaCO ₃	PET	CaCO ₃	Sand
	0.051	0.072	0.000	0.095	0.200
	Snowfall	Sand	Sand	Sand	Dispersal
	0.012	0.018	0.000	0.024	0.166
	Summer Precip.	Summer Precip.	Summer Precip.	Dispersal	Summer Precip.
	0.755	0.806	0.778	0.487	0.634
	PET	Dispersal	Dispersal	Summer Precip.	Snowfall
	0.719	0.367	0.164	0.456	0.413
	Dispersal	PET	Snowfall	Snowfall	PET
	0.322	0.124	0.021	0.195	0.268
	CaCO ₃	CaCO ₃	CaCO ₃	PET	CaCO ₃
	0.138	0.070	0.002	0.186	0.257
Wisconsin with Dispersal (east)	Snowfall	Snowfall	PET	CaCO ₃	Sand
	0.035	0.010	0.000	0.078	0.202
	Sand	Sand	Sand	Sand	Dispersal
	0.002	0.001	0.000	0.037	0.176
	Dispersal	Dispersal	Dispersal	Dispersal	Summer Precip.
	0.773	1.000	0.852	0.582	0.575
	PET	PET	Summer Precip.	Summer Precip.	PET
	0.705	0.600	0.043	0.341	0.484
	Summer Precip.	Snowfall	CaCO ₃	PET	Snowfall
	0.593	0.132	0.000	0.297	0.398
	CaCO ₃	CaCO ₃	Snowfall	CaCO ₃	CaCO ₃
	0.451	0.051	0.000	0.069	0.267
	Snowfall	Sand	PET	Snowfall	Dispersal
	0.177	0.002	0.000	0.056	0.236
	Sand	Summer Precip.	Sand	Sand	Sand
	0.041	0.000	0.000	0.033	0.234

(Continues)

**TABLE 2** (Continued)

Calibration Experiment	GAM	GLM	BRT	RF	SVM
Wisconsin with Dispersal (south)	PET	Summer Precip.	Summer Precip.	Summer Precip.	Snowfall
	0.853	0.933	0.960	0.514	0.376
	Summer Precip.	PET	Snowfall	PET	Summer Precip.
	0.731	0.703	0.026	0.441	0.370
	Dispersal	Dispersal	CaCO ₃	Dispersal	Dispersal
	0.489	0.668	0.002	0.298	0.343
	CaCO ₃	Snowfall	PET	CaCO ₃	PET
	0.281	0.135	0.000	0.084	0.277
Wisconsin without Dispersal	Snowfall	CaCO ₃	Dispersal	Snowfall	Sand
	0.140	0.082	0.000	0.071	0.252
	Sand	Sand	Sand	Sand	CaCO ₃
	0.053	0.044	0.000	0.029	0.208
	Summer Precip.	Summer Precip.	Summer Precip.	Summer Precip.	Summer Precip.
	0.851	0.961	0.953	0.623	0.474
	PET	PET	Snowfall	PET	Snowfall
	0.7051	0.3674	0.0206	0.2943	0.4467
	CaCO ₃	CaCO ₃	CaCO ₃	CaCO ₃	PET
	0.360	0.067	0.003	0.180	0.399
	Sand	Sand	PET	Snowfall	Sand
	0.059	0.034	0.000	0.095	0.316
	Snowfall	Snowfall	Sand	Sand	CaCO ₃
	0.031	0.012	0.000	0.071	0.200

TABLE 3 Goodness-of-fit for the general linear model (GLM) and residual autocovariate model (RAC) calibrations

	Goodness-of-Fit Metric	GLM	RAC
MI-MI	AUC	0.748	0.775
	TSS	0.392	0.420
WI-WI with Dispersal (east)	AUC	0.915	0.961
	TSS	0.871	0.857
WI-WI without Dispersal	AUC	0.855	0.944
	TSS	0.717	0.872

dispersal, Lake Michigan, also exerts a strong climatic effect, moderating regional temperatures and increasing downwind winter precipitation (Davis et al., 1986; Henne et al., 2007). The analyses presented here have shown that it is possible to disentangle the role of these large lake effects on the environmental and dispersal controls on *F. grandifolia* distributions. The *F. grandifolia* case study is also notable because of its well-documented record of Holocene range expansions (Bennett, 1985; Davis et al., 1986; Wang et al., 2015) and the extensive data on its distributions just prior to intensified EuroAmerican land use (Goring et al., 2016).

Whether the distribution of *F. grandifolia* and its sister species *F. sylvatica* in Europe primarily represents current environmental factors or historical legacies of slow postglacial range expansions from glacial refugia has remained a long-discussed topic (Bennett, 1985;

TABLE 4 Variable ranking for the general linear model (GLM) and residual autocovariate model (RAC) calibrations

	GLM	RAC
MI-MI	Summer Precip.	Snowfall
	Snowfall	Summer Precip.
	Sand	Sand
	PET	PET
	CaCO ₃	CaCO ₃
WI-WI with Dispersal (east)	Dispersal	Dispersal
	PET	PET
	Snowfall	Snowfall
	CaCO ₃	CaCO ₃
	Sand	Sand
WI-WI without Dispersal	Summer Precip.	Summer Precip.
	PET	PET
	CaCO ₃	Snowfall
	Sand	CaCO ₃
	Snowfall	Sand

Davis et al., 1986; Denton & Barnes, 1987; Fang & Lechowicz, 2006; Henne et al., 2007). Huntley et al. (1989) showed that *F. grandifolia* in eastern North America and *F. sylvatica* in Europe had similar

climate spaces, suggesting that both were in approximate equilibrium with climate and niche conservatism was high. Davis et al. (1986) proposed that the expansion of *F. grandifolia* into western Wisconsin was governed by the postglacial rise in temperatures but slowed by short seed dispersal distances and the Lake Michigan barrier. Since these pioneering works, a new generation of SDM models (Allouche et al., 2006; Elith & Graham, 2009) and spatial datasets of climate and historical tree distributions (Goring et al., 2016) has enabled this paper's closer look at the environmental and dispersal factors governing *F. grandifolia* distributions. Saltr  et al. (2013), also using SDMs, similarly concluded that *F. sylvatica* in Europe is determined by both dispersal limitations and environmental filtering.

These findings of dispersal limitation suggest that *F. grandifolia* (and *F. sylvatica*) may not have fully occupied its fundamental climatic niche. If so, studies that use contemporary or historic distributions of *F. grandifolia* to study species–climate relationships or trait–environment relationships (Morin, Viner, & Chuine, 2008; Svenning & Skov, 2004) may contain hidden biases with respect to conclusions drawn for *F. grandifolia* or other mesic tree taxa, such as *Tsuga canadensis* or *F. sylvatica*, for which postglacial range expansions have still been underway in recent millennia. Similarly, efforts to predict future climate-driven range shifts based only on current species distributions and derived species–environment relationships (Iverson & McKenzie, 2013; Morin et al., 2008; Svenning & Skov, 2004) may underpredict the future adaptive capacity and ranges of *F. grandifolia* and other historically dispersal-limited species.

4.2 | Differences in species–environment relationships between range-edge and core populations

A stronger sensitivity to environmental stressors in edge populations, limited gene flow within edge populations and between edge and core populations, and stronger selection for dispersal in edge populations can result in divergent responses to environmental variables between edge and core populations (Sexton et al., 2009). We found moderate to low transferability between models developed for the range-edge populations in Wisconsin and predicted to the core populations of Michigan, and vice versa, suggesting the two populations may be differentially sensitive to environmental factors and in particular, seasonality of precipitation. In Wisconsin, summer rainfall emerges as a primary environmental control on *F. grandifolia* distributions, suggesting that water availability during the growing season may be a primary control for Wisconsin populations, while both winter snowfall and summer precipitation emerge as important predictors for the Michigan core populations (Table 2). Interactions with edaphic factors (sand and CaCO_3) also varied between the core and range-margin populations.

Our findings for Michigan are consistent with those reported by Henne et al. (2007), who found a strong response to lake-effect snowfall (highest immediately downwind of Lake Michigan) on *F. grandifolia*, *Tsuga canadensis*, and *Acer saccharum* distributions in northern Michigan. Henne et al. (2007) also reported spatially varying interactions among snowfall, soil texture, and the presence of mesic tree species.

The Michigan snowbelt receives large quantities of snow (<365 mm), allowing *F. grandifolia* and other mesic taxa to establish in coarse-textured soils due to high rates of water infiltration (Henne et al., 2007). This spatially varying interaction between precipitation and edaphic factors may explain why the predictive power of SDMs fitted to Michigan populations were lower than for SDMs fitted to Wisconsin populations (Figure 5). Alternatively, the lower predictive power of the SDM models for Michigan may suggest a weaker role of environmental filtering on spatial distributions of *F. grandifolia* within its core region of abundance, consistent with theory (Sexton et al., 2009).

4.3 | Limitations and advantages of models and data and future work

While correlational SDMs are useful tools for testing hypotheses about the processes governing species distributions, they are correlational in nature and so can be influenced by choice of model type, goodness-of-fit metric, model assumptions, spatial grain and extent, quality of observational data used to fit the SDM, and failure to incorporate important explanatory factors (Elith & Graham, 2009; Guisan & Thuiller, 2005). This study attempted to reduce the limitations introduced through model selection by including five different SDMs, and the difference in SDM performance across all experiments (Figures 5 and 6, Tables 1–3) highlights the importance of assessing multiple models. Our major conclusions are insensitive to choice of goodness-of-fit metrics (Figure 5, Table 3).

This study did not attempt to model species interactions and so this remains an unevaluated control on *F. grandifolia* distributions and a potential explanation for differences between its range-edge and core populations. Various community-level models and joint species distribution models now enable simultaneous modelling of multiple species and the effect of interactions among species distributions (Clark, Nemergut, Seyednasrollah, Turner, & Zhang, 2016; Guisan & Thuiller, 2005; Nieto-Lugilde, Maguire, Blois, Williams, & Fitzpatrick, 2018; Ovaskainen, Hottola, & Siitonen, 2010). We opted not to pursue community-level modelling here in order to keep the emphasis of this paper on the relative roles of dispersal and environmental limits and because the species lists for Wisconsin and Michigan are generally similar (Burns & Honkala, 1990), making differential species interactions between the two regions less likely to be a predictor of *F. grandifolia* distributions.

Environmental factors interact with species at varying spatial scales, which exert an important influence on species distribution modelling and associated conclusions (Guisan & Thuiller, 2005). The regional extent and 8-km spatial grain used here enable a close examination of the climatic, edaphic, and dispersal controls on *F. grandifolia* distributions. However, the 8-km grain may be too coarse to capture spatially varying species–environment interactions reported at finer grains (Henne et al., 2007). A key advantage of this study is the use of historical tree data, from the period just prior to extensive EuroAmerican land use. This allows study of environmental controls and dispersal limitation for a time period when human influences, while not entirely absent, were relatively modest; Native American populations likely modified fire regimes to manage resources from fruit and mast trees

and for game management (Abrams & Nowacki, 2008; Dorney & Dorney, 1989; Muñoz, Mladenoff, Schroeder, & Williams, 2014). Future work could combine these settlement-era species distribution data with contemporary forest surveys of *F. grandifolia* to build dynamic range models (DRMs). These models include multiple sampling time points to understand the controls on species range dynamics over space and time (Schurr et al., 2012). Such data could also be augmented by palaeoecological data for earlier time periods (Bennett, 1985; Davis et al., 1986; Wang et al., 2015) and with work currently underway to build new pollen-vegetation models to better quantify past changes in forest composition (Dawson et al., in revision). A next step forward is to more fully integrate the historical datasets analysed here with both contemporary data on tree distributions, genetic evidence, and palaeoecological observations data in order to understand the environmental and ecological controls on species distributions and dynamics.

5 | CONCLUSIONS

When assessing species range dynamics, one fundamental question is whether the species is in equilibrium with its environment, or whether dispersal limitations persist. We assessed that question here using *F. grandifolia* populations in Michigan, where dispersal limitations should be minor or absent, and Wisconsin, where we test for disequilibrium caused by dispersal limitation. These experiments suggest that *F. grandifolia* is limited by both dispersal and environmental variables in Wisconsin. Evidence for dispersal limitation includes the improved predictive ability of five SDMs when distance to *F. grandifolia* entry into Wisconsin is included as a predictive variable, the selection of dispersal as a strong predictive variable of *F. grandifolia* distributions in Wisconsin, and the minimal signal of SAC in the models that include a dispersal proxy. Conversely, evidence for environmental filtering includes low goodness-of-fit for SDMs that exclude environmental predictors and high rankings of individual environmental variables. Together, this evidence leads us to conclude that dispersal and environmental factors together determined the range margin of *F. grandifolia* just prior to EuroAmerican settlement, highlighting the importance of evaluating dispersal limitations when assessing species range limits. In Michigan, environmental factors more weakly predicted *F. grandifolia* ranges, suggesting a stronger influence of environmental factors for the range-edge populations of Wisconsin and the possible effects of spatially varying interactions between precipitation and soil texture at finer spatial grains. Future insights into the environmental, ecological, and anthropogenic controls on the distributions of *F. grandifolia* and other tree species can be gained by linking contemporary, historical, and palaeoecological data on species distributions with dynamic biogeographical models.

ACKNOWLEDGMENTS

This contribution to the PalEON (Palaeoecological Observatory Network) was supported by the University of Wisconsin-Madison Holstrom Environmental Scholarship and the National Science

Foundation (EF-1065656, EF-1241868). We thank David Mladenoff for his guidance and support. The witness tree datasets were assembled by the Settlement Vegetation working group of PalEON with contributions by Charles Cogbill, Jason McLachlan, Chris Paciorek, and Jody Peters. For creating and making available the PRISM and SSURGO datasets, we acknowledge the PRISM Climate Group and National Resources Conservation Service, respectively.

DATA ACCESSIBILITY

The Goring et al. (2016) *F. grandifolia* pre-EuroAmerican settlement distribution data can be acquired from https://figshare.com/articles/S1_RawCode_Data_zip/4024860/1. Soil property datasets are available at the Natural Resource Conservation Service SSURGO database (<https://datagateway.nrcs.usda.gov/>). Historic climate data can be accessed from the PRISM Climate Group (<http://prism.oregonstate.edu/historical/>).

Digital elevation models are available through the Wisconsin Department of Natural Resources (https://p.widencdn.net/tebltdt/DEM_30m) and the Michigan Department of Technology, Management, and Budget (http://www.dnr.state.mi.us/spatialdatalibrary/sd12/Topography/DEM/_Statewide/Michigan_DEM_250k_90m.exe) for Wisconsin and Michigan, respectively.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

How to cite this article: Seeley M, Goring S, Williams JW.

Assessing the environmental and dispersal controls on *Fagus grandifolia* distributions in the Great Lakes region. *J Biogeogr*. 2019;46:405–419. <https://doi.org/10.1111/jbi.13491>